<u>Sequence</u> : $f : \mathbb{N} \to \mathbb{R}$.

$\{a_n\}$

$a_1, a_2, a_3, \dots, a_n, a_{n+1}, \dots \to$ "$\underline{a}$"

1st term of the sequence

"n+1"th.

nth term of the sequence

$$\lim_{n\to\infty} a_n = a$$

If i have no such "a" Limit does not exists.

**Ex.:** $a_n = 1/n$.

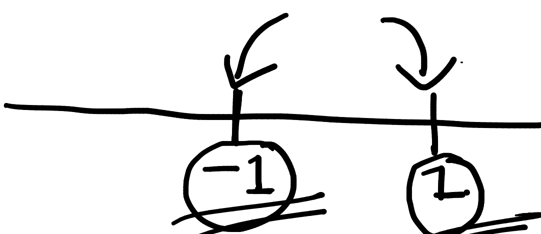
$1, \; 1/2, \; 1/3, \; 1/4, \; \dots$ "$\underline{0}$"

$$\lim_{n\to\infty} 1/n = 0$$

**Ex:** $\underline{a_n = (-1)^n}$.

$-1, \; 1, \; -1, \; 1, \; \dots$

$\lim_{n\to\infty} a_n$ - D.N.E.



**Ex:**

$$a_n = \frac{n^2+7n+5}{2n^2+9n+1} = \frac{1 + \underbrace{7/n}_{\to 0} + \underbrace{5/n^2}_{\to 0}}{2 + \underbrace{9/n}_{\to 0} + \underbrace{1/n^2}_{\to 0}} \quad \begin{matrix} \to \underline{1}. \\ \to 2. \end{matrix}$$

Mathematics for Data Science - 1
**REVISION SESSION- Weeks: 7,8**

1. Find the limit of the sequence $a_n = \sqrt{2n+1} - \sqrt{2n}$.

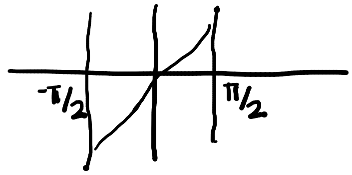
$$a_n = \sqrt{2n+1} - \sqrt{2n} \cdot \frac{(\sqrt{2n+1} + \sqrt{2n})}{\sqrt{2n+1} + \sqrt{2n}}$$

$$= \frac{2n+1 - 2n}{\sqrt{2n+1} + \sqrt{2n}} = \frac{1}{\sqrt{2n+1} + \sqrt{2n}} \to \infty$$

as $\underline{\underline{n \to \infty}}$: $\underline{\underline{a_n \to 0}}$

$\underline{\underline{a_n = n}}$

$n \to \infty$ $\quad a_n \to \infty \to$ <u>divergent</u>

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12

2. Find the limit of the sequence $\{a_n\}$ such that

$$a_n = \begin{cases} \dfrac{10n + 1 + \sin(n)}{n} & \text{if } n \text{ is odd} \\ \dfrac{10n - 1 + \cos(n)}{n} & \text{if } n \text{ is even} \end{cases}$$

$n \to \infty$

$a_n \to 2?$

$a_1 = \dfrac{10 + 1 + \sin 1}{1}$, $a_2 = \dfrac{20 - 1 + \cos(2)}{2}$, - - - - - - -

$n$-is odd & $n \to \infty$ : $\dfrac{10n + 1 + \sin(n)}{n} = 10 + \underbrace{\dfrac{1}{n}}_{\to 0} + \dfrac{\sin(n)}{n} \to 10$

$n$ - natural number

$\dfrac{\sin(n)}{n}$

$\sin(n)$

$0 \leftarrow \dfrac{-1}{n} \leq \dfrac{\sin(n)}{n} \leq \dfrac{1}{n} \to 0$

$\dfrac{\sin(n)}{n} \to 0$

$\dfrac{-M}{n} < \dfrac{f(n)}{n} < \dfrac{M}{n}$

$\dfrac{\cos(n)}{n} \to 0$

If $n$ is even; $a_n \to 10$

• $\lim\limits_{n\to\infty} (a_n + b_n)$

$= \lim\limits_{n\to\infty} a_n +$

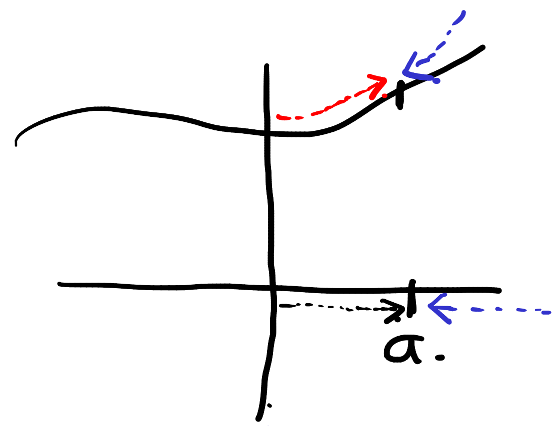
$\lim\limits_{n\to\infty} (n)^{1/n} = 1$

**Limit of a Function:**

- If $x$ approaches to $a$ from the right side of $a$, and the value of $f$ approaches to $l_R$, then it is denoted by $\lim\limits_{x\to a+} = l_R$.
- If $x$ approaches to $a$ from the left side of $a$, and the value of $f$ approaches to $l_L$, then it is denoted by $\lim\limits_{x\to a-} = l_L$.
- If $l_R = l_L$, then the limit at the point $a$ exists and it's value is $l_R = l_L = l$ (say).

$$\lim_{x\to a-} f(x) = \lim_{x\to a+} f(x) = \lim_{x\to a} f(x) = l$$

$f : \mathbb{R} \longrightarrow \mathbb{R}.$



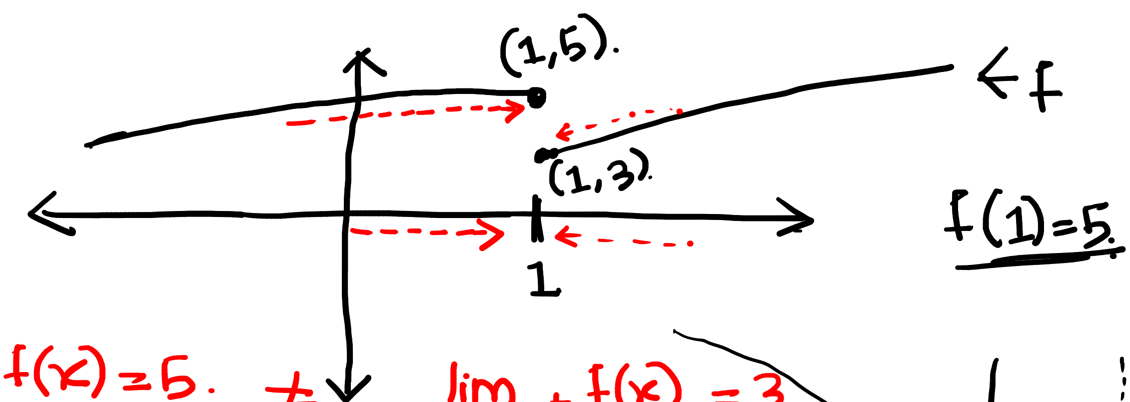

$\lim\limits_{x\to a} f(x).$

— L.H.L

$\lim\limits_{x\to a^-} f(x) = l_L.$

$\lim\limits_{x\to a^+} f(x) = l_R.$

If $l_L = l_R \Rightarrow$ at "$x=a$" exists

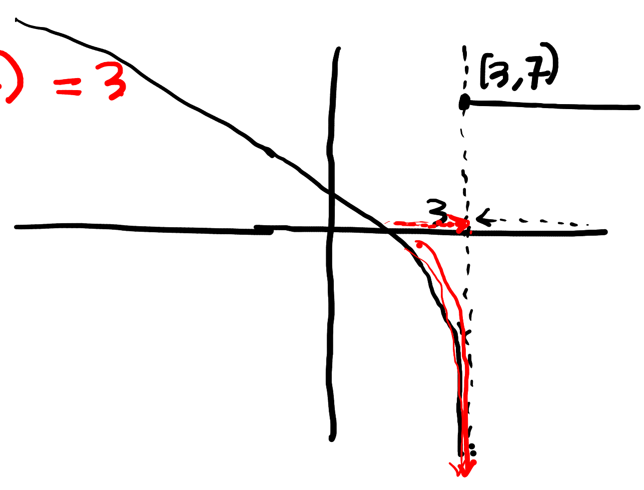



$f(1)=5.$

$\lim\limits_{x\to 1^-} f(x) = 5. \neq \lim\limits_{x\to 1^+} f(x) = 3$

$\Rightarrow$ L.D.E.

$+\infty, -\infty$



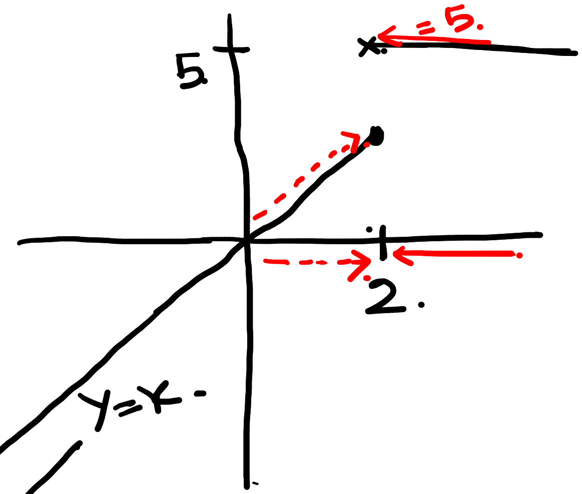

- $\lim\limits_{x\to 3^-} f(x) = -\infty$ (L.D.E). $\Rightarrow$
- $\lim\limits_{x\to 3^+} f(x) = 7.$

3. Define a function

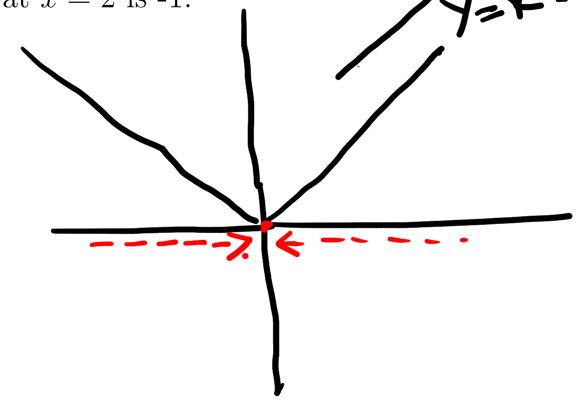
$$f(x) = \begin{cases} x & \text{if } x \leq 2 \\ 5 & \text{if } x > 2 \end{cases}$$

Which of the following option(s) is(are) true?

- ◯ Option 1: Limit exists at $x = 2$.
- ◯ Option 2: Right limit exists at $x = 2$.
- ◯ Option 3: Left limit at $x = 2$ is 2.
- ◯ Option 4: Right limit at $x = 2$ is -1.




$f(x) = |x|.$

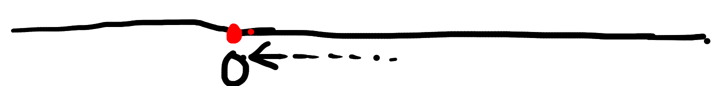


4. Which of the following option(s) is(are) true?

- ◯ Option 1: $\lim\limits_{x \to -1} \dfrac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$
- ◯ Option 2: $\lim\limits_{x \to 0} \dfrac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$
- ◯ Option 3: $\lim\limits_{x \to 2} (x^3 + 4x^2 - 6x - 7) = 5$
- ◯ Option 4: $\lim\limits_{x \to 3} \dfrac{x^2 - 6x + 9}{x - 3} = 1$

$f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}.$

$f(x) = x.$




$x = 0$

$\lim\limits_{x \to 0} f(x) = l.$

$\lim\limits_{x \to 0} f(x) = \cancel{f(0)}.$

$f(a)$

$a \in Dom(f).$

$a$ $\quad a \in Dom(f)$

**(Sandwich principle):** If $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = L$, and $h(x)$ is a function such that $f(x) \le h(x) \le g(x)$, then $\lim_{x\to a} h(x) = L$

$L \leftarrow f(x) \le h(x) \le g(x) \rightarrow L$

$L$.

$\cos 1/x$ - undefined

← not defined at "0"

$x \neq 0$

5. Let $f(x) = x^2 \cos\frac{1}{x}$. Find $\lim_{x\to 0} f(x)$.

$\lim_{x\to 0} f(x) = 0 = f(0)$

$-1 \le \cos 1/x \le 1$

$x^2 > 0$

$-1 \le \cos 1/x$ , $\cos 1/x \le 1$.

$\alpha < 0$

$\alpha \cos 1/x \le -\alpha$.

$\alpha \le \alpha \cos 1/x$

$\alpha \le \alpha \cos 1/x \le -\alpha$.

$x \cos 1/x$

0

$x$-is negative

$x > 0$

**Continuity at a point:**

A function $f$ is said to be continuous at a point $x = a$ in its domain, if

1) f(a) exists; i.e value of $f(x)$ at $x = a$ exists.
2) $\lim_{x\to a} f(x)$ exists; i.e both left and right limits exist and are equal.
3) $\lim_{x\to a} f(x) = f(a)$.

Equivalently,

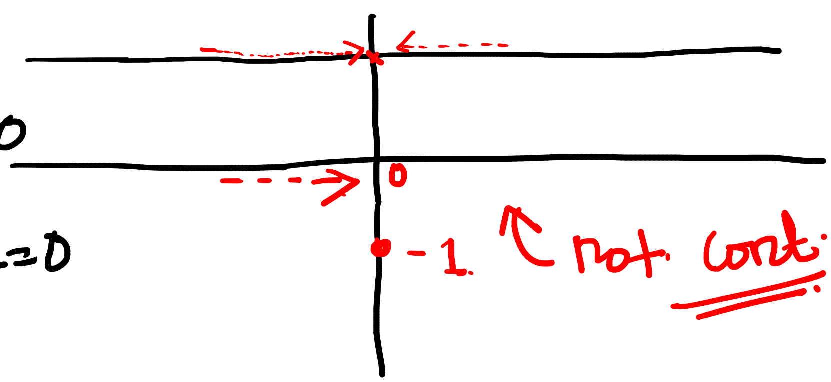
we can say that $f$ is **continuous** at $x = a$, if for all $\{x_n\} \to a$ implies that $\{f(x_n)\} \to f(a)$.

- $f : \mathbb{R} \to \mathbb{R}$.

  L.D.N.E at $x=a$. $\Rightarrow$ f-is not cont at $x=a$

$\lim_{x\to a} f(x) = f(a) \Rightarrow$ f-is cont.

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ -1 & \text{if } x = 0 \end{cases}$$




$\lim_{x\to 0^-} f(x) = 1 = \lim_{x\to 0^+} f(x) \neq f(0)$.

6. Check the continuity of the function

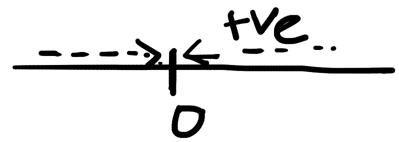
$x$ $\quad$ $x = 1$ $\quad$ $\frac{x^2}{|x|}$

$$f(x) = \begin{cases} \frac{x^2}{|x|} & \text{if } x \neq \cancel{0}\,1, \\ 0 & \text{if } x = \cancel{0}\,1 \end{cases}$$

$f(0) = 0$

~~at $x = 0$~~

$$\lim_{x \to 0^-} \frac{x^2}{|x|} = \lim_{x \to 0^-} \frac{x^2}{-x}$$




$$= \lim_{x \to 0^-} -x = 0 \quad \text{— L.H.L}$$

If $x$ – ve

$|x| = -x$.

$$\lim_{x \to 0^+} \frac{x^2}{x} = \lim_{x \to 0^+} x = 0 \quad \text{— R.H.L}$$

• $x$ – +ve

$|x| = x$

$x$ – –ve

$|x| = -x \geq 0$

$x$ – +ve

$x = -2$

$|-2| = -(-2) = 2$

$|-2| = -(-2) = 2$

$|x| = x$

$x = 2$

$|2| = 2$

7. Define a function $f$ as follows:

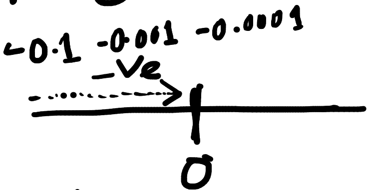
$$f(x) = \begin{cases} \dfrac{1}{e^{\frac{1}{x}}+1} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$ $\leftarrow$ L.D.E.

Which of the following option(s) is(are) true?

- ○ Option 1: $\lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$
- ○ Option 2: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x)$
- ~~○ Option 3: $f$ is a bounded function on $\mathbb{R}$.~~
- ○ Option 4: $f$ is continuous at $x = 0$.

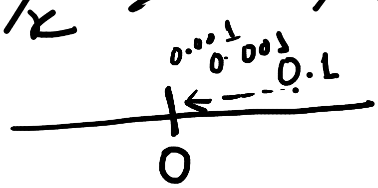
$$f(x) = \begin{cases} \dfrac{1}{e^{1/x}+1} & \text{If } x \neq 0 \\ 0 & \text{If } x = 0 \end{cases}$$

$e^{-\infty} \to 0$




**L.H.L**

$x \to 0^-,\ 1/x \to -\infty,\ e^{1/x} \to 0,\ \dfrac{1}{e^{1/x}+1} \to 1$




$\neq$

**R.H.L**

$x \to 0^+,\ 1/x \to +\infty,\ e^{1/x} \to \infty,\ \dfrac{1}{e^{1/x}+1} \to 0$

$1+e^{1/x} \to \infty$

$e^{x} > 0 \Rightarrow e^{1/x} > 0$

$\Rightarrow 1+e^{1/x} \geq 1$

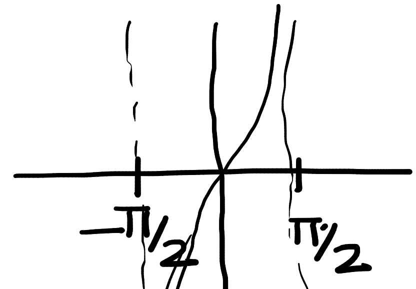
$\Rightarrow \boxed{0 < \dfrac{1}{1+e^{1/x}} < 1}$

7. Define a function $f$ as follows:

$$f(x) = \begin{cases} \dfrac{1}{e^{\frac{1}{x}}+1} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

Which of the following option(s) is(are) true?

- ○ Option 1: $\lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$
- ○ Option 2: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x)$
- ○ Option 3: $f$ is a bounded function on $\mathbb{R}$.
- ○ Option 4: $f$ is continuous at $x = 0$.




$\leftarrow \pi/2$

$f(x) = \tan x$

$e^x > 0 \quad \forall x \in \mathbb{R}\setminus\{0\}$

$\Rightarrow e^{1/x} > 0$

$\Rightarrow 1 + e^{1/x} > 1$

$\Rightarrow 0 < 1/(1+e^{1/x}) < 1$ If $x \neq 0$

$\to -1$

$f(x) \in [0, 1) \quad \forall x \in \mathbb{R}$

$f(x) = \frac{x}{x+1}$

$0$

If $x = 0$

$x^2 + 1 > x^2$

$0 \leq x^2/(x^2+1) < 1$

$f(x) = \frac{x^2}{x^2+1}$

- $\exists$ non-constant cont $f : \mathbb{R} \to \mathbb{R}$ which is bounded?

Ex: $f : \mathbb{R} \to \mathbb{R}$.

$f(x) = \sin x \to \cos x$.

$-1 \leq \sin x \leq 1$

7. Define a function $f$ as follows:

$$f(x) = \begin{cases} \dfrac{1}{e^{\frac{1}{x}} + 1} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

Which of the following option(s) is(are) true?

- ○ Option 1: $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$
- ○ Option 2: $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x)$
- ○ Option 3: $f$ is a bounded function on $\mathbb{R}$.
- ○ Option 4: $f$ is continuous at $x = 0$.

• $\exists$ unbounded cont. function from $(2, 3]$ to $\mathbb{R}$.

$f : (2, 3] \longrightarrow \mathbb{R}$

$\cancel{f : [2,3] \to \mathbb{R}}$

$\cancel{3 \le f(x) = x+1 \le 4}$

$2 \notin \text{Dom}(f)$

$f(x) = \dfrac{x}{x-2} \to \infty$ as $x \to 2$

$\downarrow 0$

$\uparrow$ unbounded + cont.

**Differentiability:**

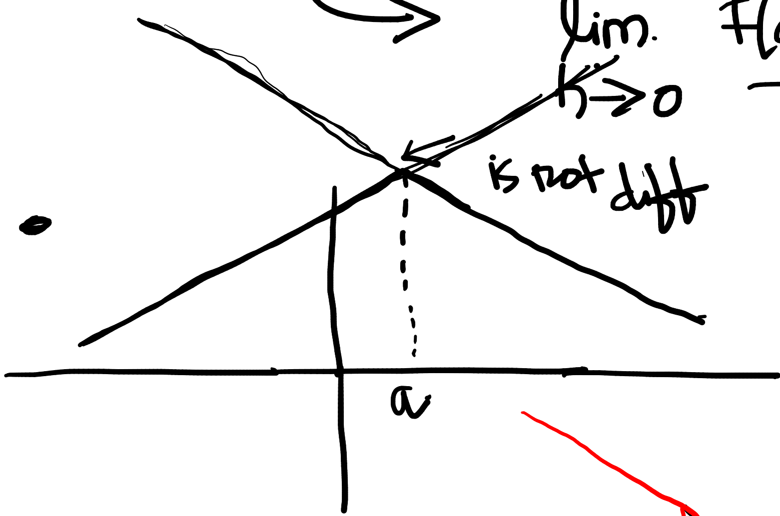
- **Differentiability:** Let $f$ be a function defined on an interval around $a$. Then $f$ is differentiable at a if $\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ exists.
- **Differentiability implies continuity:** If $f$ is differentiable at $a$, then it is continuous at $a$.
- Checking differentiabilty by looking of the graph of a function: A function is not differentiable at a point if there is a sharp corner at that point, i.e., there is no tangent line, or there is a vertical tangent line.
- **Equation of the tangent at a point on a curve:** Let $f(x)$ be differentiable at the point $a$. Then the tangent to $f$ at $a$ exists and is given by

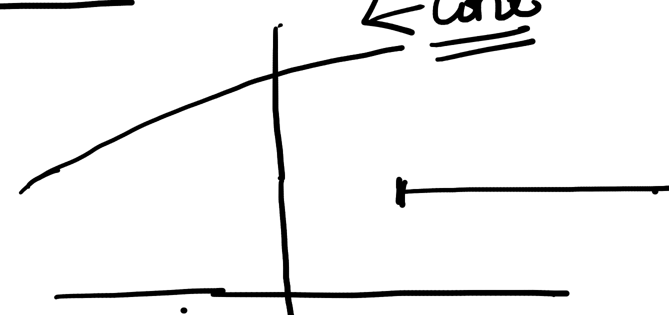
$$y = f'(a)(x-a) + f(a).$$

- $f: \mathbb{R} \longrightarrow \mathbb{R}$

  ↳ at $\underline{x=a}$

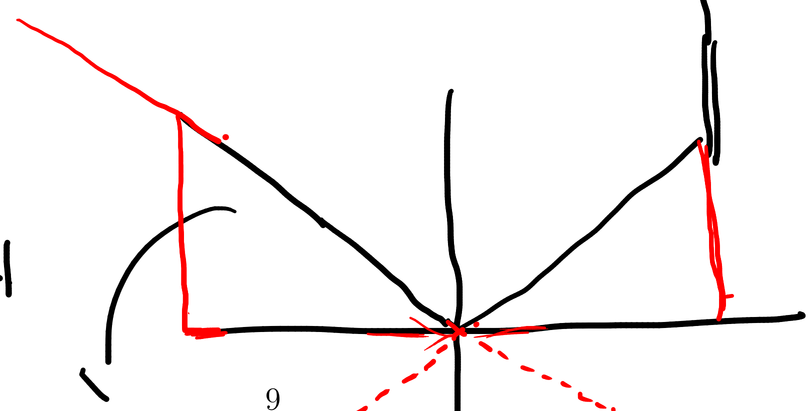
  ↳ $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \to .$

- 


[-3, 3]

- $f(x) = |x|$



cont $x=0$ — not diff at $x=0$

8. Define a function $f$ as follows:

$$f(x) = \begin{cases} kx^2 + l & \text{if } x \leq 1 \\ lx^2 + kx + m & \text{if } x > 1 \end{cases}$$

If $f$ is continuous and differentible at $x = 1$ then the value of $k - 2l + m$ is

$\Rightarrow$ $k + l = l + k + m \Rightarrow m = 0$

$$f'(x) = \begin{cases} 2kx & x \leq 1 \\ 2lx + k & x > 1 \end{cases}$$

$\Rightarrow 2k = 2l + k \Rightarrow 2l - k = 0$

$f'(x) : \mathbb{R} \to \mathbb{R}$

$f$ - diff on $\mathbb{R}$

$f'(x)$ $\forall x \in \mathbb{R}$

9. Choose the set of correct options.

$e^x$

$f \neq 0 \to 1/f$ is diff

- ○ ~~The derivative of the function $f(x) = \sin(2\cos 3x)$ is $f'(x) = 6\sin(3x)\cos(2\cos 3x)$.~~
- ○ If $h(x) = x^2 + \sin(\pi x) + e^x$ is the derivative of the function $f$ then the value of $\lim_{m\to 0} \frac{f(2+m) - f(2)}{m}$ is $4 + e^2$.
- ○ The function $f(x) = \frac{x}{e^x}$ is differentiable on $\mathbb{R}$.
- ○ If product of two functions $f$ and $g$ is differentiable, then both $f$ and $g$ are differentiable.

$f'(x)=h(x)$

$x \times 1/e^x$

$e^x \neq 0$

$1/e^x$

If $f$ & $g$ diff then $fg$ is

$$\lim_{m\to 0}\frac{f(2+m)-f(2)}{m} = f'(2) = h(2) = 4+\sin(2\pi)+e^2 = 4+e^2$$

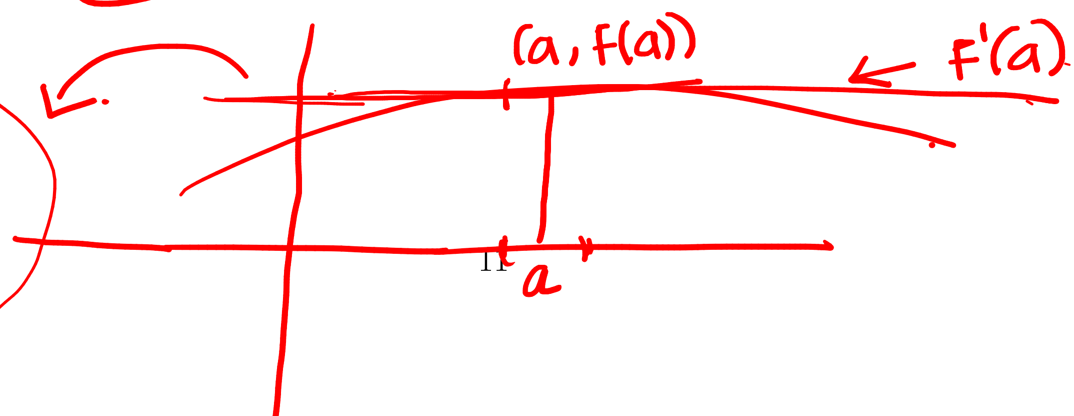
$$f(x) = \sin(2\cos 3x)$$

$$f'(x) = \cos(2\cos 3x) \times 2 \times -\sin(3x) \times 3$$

$f(x) = x$ $\quad g(x) = |x|$ $\quad f \times g = x|x|$

$\cancel{f(x)} = |x|+1$ $\quad \cancel{g(x)} = \frac{1}{|x|+1}$ $\quad fg = 1$

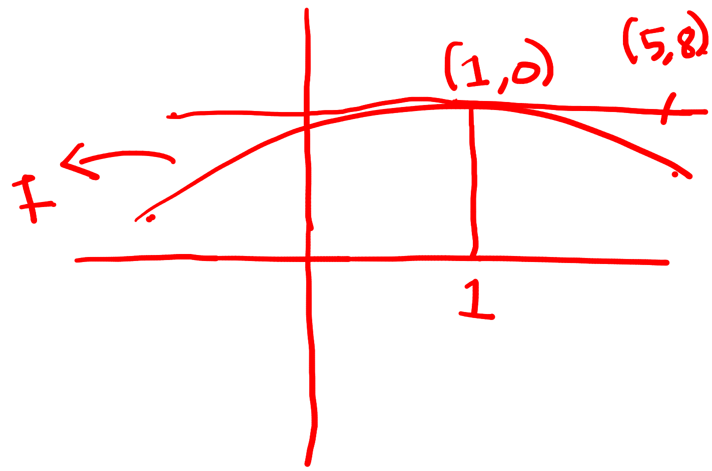
$$f(x) = \begin{cases} x^2 & \text{If } x \geq 0 \\ -x^2 & \text{If } x < 0 \end{cases}$$




$y = f'(a)(x-a) + f(a)$

10. Let $f(x)$ be differentiable at $x = 1$ and $f$ be represented by the curve $C$. If a line which passes through the point $(5, 8)$, is the tangent to the curve $C$ at the point $(1, 0)$, then find the value of $f'(1)$.

$f'(1)$




$f(x) = x|x|$

$\hookrightarrow$ $x > 0$ | $x < 0$

$|x| = x$ | $|x| = -x$

$$f(x) = \begin{cases} x^2 & \text{If } x \geq 0 \\ -x^2 & \text{If } x \leq 0 \end{cases}$$

$q(t) \rightarrow$ at $t = 3 \rightarrow y = 2x + 3$

$\hookrightarrow L_p(t) = 2t + 3$

$L_p(t) = Ax + B$

$A = 2, B = 3$

$3/2 \times$ -ve -ve

$q(t) = |t(t-2)(t-4)|$

$|x| = x$

$|x| = -x$

0 2 4

$$\boxed{q(t) = t(t-2)(t-4)}$$

0 3/2 2 4

$q'(t) \rightarrow q'(3/2)$

$q'(3/2) = 3 \times 9/4 - 12 \times 3/2$

$= 27/4 - 18$

$g'(t) = (t-2)(t-4) + t(t-4) + (t-2)t = 3t^2 - 12t + 8$

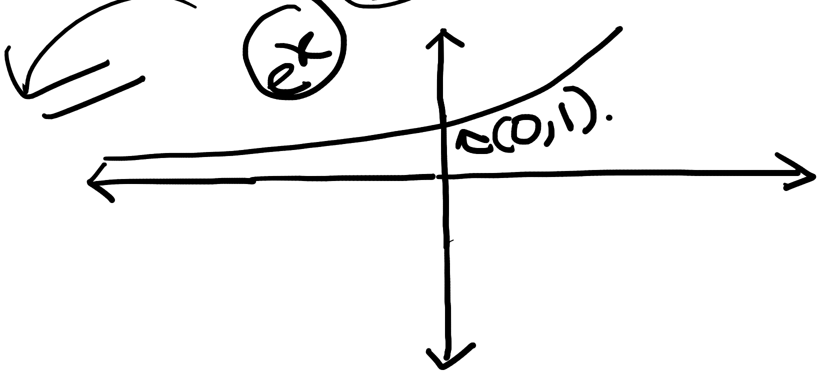
10. Let $f(x)$ be differentiable at $x = 1$ and $f$ be represented by the curve $C$. If a line which passes through the point $(5, 8)$, is the tangent to the curve $C$ at the point $(1, 0)$, then find the value of $f'(1)$.

$|x|$

$|x+1|$ diff.

$-1$

$e^x$ $e^x$




$a \neq e$

$a^x \quad a > 1$